

ANSWER KEY — Integration Lesson 29 Test: Integral Properties & Symmetry

For the grader · full working shown · give credit for correct method with small arithmetic slips

1. Given $\int_0^8 f \, dx = 15$ and $\int_0^3 f \, dx = 6$, find $\int_3^8 f \, dx$.

Work: splitting rule: $\int_0^8 = \int_0^3 + \int_3^8$, so $15 = 6 + \int_3^8 f \, dx \rightarrow \int_3^8 f \, dx = 15 - 6$. **Answer: 9.**

2. Given $\int_2^9 f \, dx = 4$, find (a) $\int_9^2 f \, dx$ and (b) $\int_5^5 f \, dx$.

Work: (a) reversal rule (swapped limits flip the sign): $\int_9^2 f \, dx = -\int_2^9 f \, dx = -4$. (b) zero-width rule: equal limits $\rightarrow$ region of width 0 $\rightarrow$ area 0. **Answer: (a) -4 (b) 0.**

3. Given $\int_1^6 f \, dx = 7$ and $\int_1^6 g \, dx = 2$, find $\int_1^6 (f + 4g) \, dx$.

Work: linearity: $\int(f + 4g) = \int f + 4 \cdot \int g = 7 + 4 \cdot 2 = 7 + 8$. **Answer: 15.**

4. Classify: (a) $x^6 + 3$ (b) $x^5 - x$ (c) $x^3 + 1$.

Work: (a) $f(-x) = (-x)^6 + 3 = x^6 + 3 = f(x) \rightarrow$ **even**. (b) $f(-x) = (-x)^5 - (-x) = -x^5 + x = -(x^5 - x) = -f(x) \rightarrow$ **odd**. (c) $f(-x) = -x^3 + 1$; this is neither $f(x) = x^3 + 1$ nor $-f(x) = -x^3 - 1 \rightarrow$ **neither** (the constant +1 is even, the x^3 is odd; the mix wrecks both symmetries). **Answer: (a) even (b) odd (c) neither.**

5. Evaluate $\int_{-3}^3 x^3 \, dx$ using symmetry, then verify with the FTC.

Work: x^3 is odd and $[-3, 3]$ is symmetric $\rightarrow$ integral = 0 (left area cancels right area). FTC check: antiderivative $x^4/4$; at 3: $81/4$; at -3: $(-3)^4/4 = 81/4$; $81/4 - 81/4 = 0 \checkmark$. **Answer: 0.**

6. Evaluate $\int_{-1}^1 x^2 \, dx$ using the even shortcut, then verify by the FTC straight across.

Work: x^2 is even $\rightarrow 2 \cdot \int_0^1 x^2 \, dx = 2 \cdot [x^3/3]_0^1 = 2 \cdot (1/3) = 2/3$. Straight across: $[x^3/3]_{-1}^1 = 1/3 - (-1/3) = 2/3 \checkmark$. **Answer: 2/3.**

7. Evaluate $\int_{-2}^2 (x^5 + 3) \, dx$ by splitting into terms.

Work: x^5 is odd on the symmetric interval $[-2, 2] \rightarrow$ contributes 0. The constant 3 makes a rectangle: $3 \times (\text{width } 4) = 12$. Total = 0 + 12. **Answer: 12.** (Full-FTC check: $[x^6/6 + 3x]_{-2}^2 = (32/3 + 6) - (32/3 - 6) = 12 \checkmark$.)

8. g is EVEN and $\int_0^5 g \, dx = 9$. Find (a) $\int_{-5}^5 g \, dx$ and (b) $\int_{-5}^0 g \, dx$.

Work: (a) even shortcut: $2 \cdot 9 = 18$. (b) the left half mirrors the right half exactly, so $\int_{-5}^0 g \, dx = \int_0^5 g \, dx = 9$. (Consistency: $9 + 9 = 18$ ✓.) **Answer: (a) 18 (b) 9.**

9. f is ODD and $\int_0^4 f \, dx = 3$. Find (a) $\int_{-4}^4 f \, dx$, (b) $\int_{-4}^0 f \, dx$, (c) the average value of f on $[-4, 4]$.

Work: (a) odd on a symmetric interval $\rightarrow 0$. (b) splitting: $0 = \int_{-4}^0 + \int_0^4 = \int_{-4}^0 + 3 \rightarrow \int_{-4}^0 f \, dx = -3$ (the left half is the negative mirror of the right half). (c) average = integral $\div$ width = $0 \div 8 = 0$. **Answer: (a) 0 (b) -3 (c) 0.**

10. A student writes: " $\int_{-2}^2 x^3 \, dx = 2 \cdot \int_0^2 x^3 \, dx = 2 \cdot 4 = 8$." Explain the error, give the correct value, verify by FTC.

Work: the "double the right half" shortcut is only for EVEN functions. x^3 is ODD, so the left half is the *negative* of the right half and they cancel. Correct value: 0. FTC check: $[x^4/4]_{-2}^2 = 16/4 - 16/4 = 4 - 4 = 0$ ✓. (The student's $\int_0^2 x^3 \, dx = 4$ was correct; only the doubling was illegal.) **Answer: correct value is 0.**